

AI-BFSI Internship Project Documentation

Functional Features – Subscription Waste Detector

1. Introduction

Functional features define the specific operations and services that a software system provides to its users. They represent the core behavior of the system and form the foundation for implementation, testing, and validation.

The **Subscription Waste Detector** was developed as an AI-assisted financial analysis application that helps users identify recurring subscription expenses, visualize monthly subscription spending, receive renewal reminders, and obtain AI-powered recommendations. The implemented features focus on practical financial analysis while maintaining a simple and user-friendly interface.

2. Functional Objectives

- Accept user expense records for analysis
- Detect recurring subscription payments
- Display subscription information through an interactive dashboard
- Visualize monthly subscription spending
- Generate AI-powered financial recommendations
- Improve financial awareness regarding recurring subscriptions

3. Functional Feature Overview

Feature ID	Functional Feature	Status
FF-01	CSV File Upload	Implemented
FF-02	Excel File Upload	Implemented
FF-03	Manual Expense Entry	Implemented
FF-04	Expense Data Processing	Implemented
FF-05	Subscription Detection	Implemented
FF-06	Subscription Listing	Implemented
FF-07	Monthly Spending Analysis	Implemented
FF-08	Interactive Dashboard	Implemented
FF-09	Plotly Data Visualization	Implemented
FF-10	Renewal Reminder	Implemented
FF-11	AI-Powered Recommendations (Groq API)	Implemented

4. Functional Features

FF-01: CSV File Upload

Users can upload expense records in CSV format. Data is validated and prepared for analysis.

FF-02: Excel File Upload

Microsoft Excel files are supported via OpenPyXL, enabling spreadsheet import for subscription analysis.

FF-03: Manual Expense Entry

Users can manually enter expense details (description, category, amount, renewal date) when files are unavailable.

FF-04: Expense Data Processing

Python + Pandas process uploaded or manually entered records, organizing them into structured datasets.

FF-05: Subscription Detection

Recurring transactions are identified and grouped, with recurring costs calculated.

FF-06: Subscription Listing

Detected subscriptions are displayed with name, amount, category, and renewal date.

FF-07: Monthly Spending Analysis

Aggregates subscription costs to compute total monthly expenditure.

FF-08: Interactive Dashboard

Streamlit dashboard presents summaries, spending overviews, reminders, and AI recommendations.

FF-09: Plotly Data Visualization

Interactive charts visualize spending patterns, categories, and comparisons.

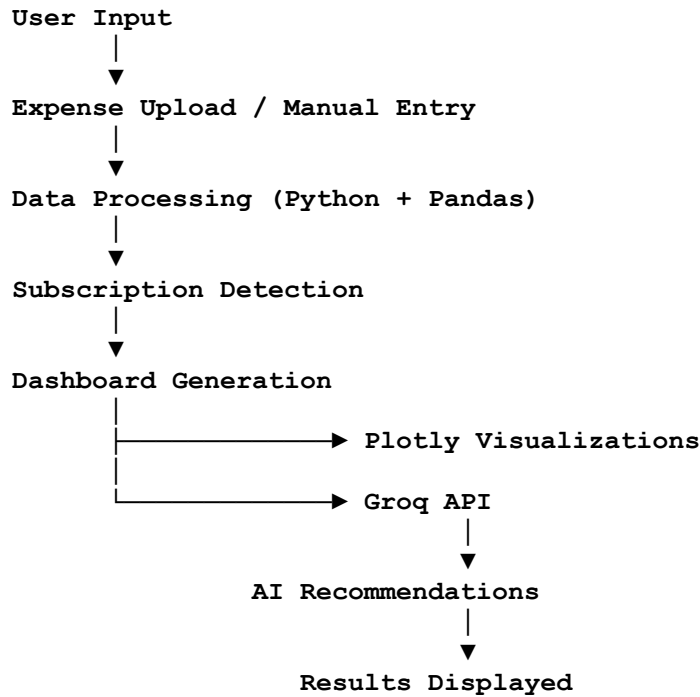
FF-10: Renewal Reminder

Upcoming subscription renewals are highlighted within the dashboard.

FF-11: AI-Powered Recommendations

Groq API generates personalized cost-saving suggestions. **Note:** The system does not train its own AI model; it uses Groq API inference.

5. Functional Workflow



6. Functional Dependencies

Feature	Depends On
Subscription Detection	Expense Data Processing
Dashboard	Subscription Detection
Plotly Charts	Processed Expense Data
AI Recommendations	Subscription Summary
Renewal Reminder	Renewal Date Information

7. Functional Validation

Validation confirmed that:

- CSV/Excel uploads work correctly
- Manual entries are accepted
- Subscription detection identifies recurring expenses
- Dashboard displays summaries and reminders
- Plotly charts render correctly
- Groq API generates recommendations

8. Current Functional Scope

Included

- CSV upload
- Excel upload

- Manual entry
- Subscription detection
- Subscription listing
- Dashboard
- Plotly charts
- Renewal reminders
- AI recommendations

Excluded

- Authentication
- Database storage
- Bank API integration
- Automatic synchronization
- Email/SMS notifications
- Mobile app
- Multi-user support
- Investment/tax/budget management

9. Conclusion

The Subscription Waste Detector implements a focused set of functional features that directly address the challenge of managing recurring subscription expenses. By combining expense import, subscription detection, visualization, renewal reminders, and AI-powered recommendations, the application delivers meaningful financial insights while remaining simple, maintainable, and aligned with the defined project scope.